Unit VI

6 Web Application Deployment

6.1 : AWS Cloud

Q.1 What is cloud computing ? [SPPU : Dec.-22, Marks 3]

Ans. : • Cloud computing is one of the most usable services in IT services where we can use this service for various operations.

- We can deliver the IT infrastructure such as servers, storage, database, networking, software, analytics and intelligence over the internet which is the cloud.
- It will make faster accessibility of the IT resources for the development and deployment of any kind of project.
- It helps to lower your operating costs and run infrastructure more efficiently and scale as your business needs change.

Q.2 What are the benefits of cloud computing ?

☞ [SPPU : Dec.-22, Marks 3]

Ans. : • Cost

- Performance speed
- Productivity
- Reliability
- Security

Global scale

Q.3 What are the types of cloud computing ?

Ans. : • IaaS - Infrastructure as a service

- PaaS - Platform as a service

- SaaS - Software as a service
- Serverless computing

Q.4 What is AWS cloud ? List different services provided by it.

[SPPU : June-22, Dec.-22, Marks 6]

Ans. : • AWS is Amazon Web Service which is a cloud platform offering over 200 fully-featured services from data centers globally.

- It is a leading cloud computing platform.
- With AWS cloud we can deliver the projects on cloud IT infrastructure.
- It has more functionality for storage, delivery, networking, database, software, etc.
- It is the most secure cloud platform as compared to other platform providers which are very flexible to end-users.
- AWS has the largest community of customers and developers for support.

AWS Services

- As AWS provides offers over 200 fully featured services which include storage, database, networking, software, etc. there are some important services that we should know about it.
  - o AWS compute services
  - o Storage
  - o Database services
  - o Security services
  - o Analytics
  - o Developer tools
  - o Deployment and management
  - o Mobile services
  - o Internet of Things (IoT)
  - o Migration

6.2 : AWS Elastic Compute

Q.5 What is EC2 service ?

Ans. :
- EC2 is a virtual machine with an operating system and hardware component that you want to use.
- AWS allows us to run various virtual computers and manage them with single hardware.
- EC2 is one of the most used primary services from the AWS ecosystem.
- EC2 enables on-demand, scalable computing capacity in the AWS cloud.
- An AWS EC2 instance eliminates the up-front investment for hardware, so there is no need to maintain any rented hardware.
- EC2 enables you to build and run applications faster.
- You can use EC2 in AWS to launch as many virtual servers as you need.

Q.6 What are the EC2 types ?

Ans. :
- General purpose
- Compute-optimized
- Memory-optimized
- Accelerated computing
- Storage optimized

Q.7 What is PuTTY ? **[SPPU : Dec.-22, Marks 2]**

Ans. :
- PuTTY is a free and open-source terminal emulator, serial console and network file transfer application. It supports several network protocols, including SCP, SSH, Telnet, rlogin, and raw socket connection. It can also connect to a serial port.

Q.8 How to connect EC2 instance with PuTTY ?

[SPPU : Dec.-22, Marks 3]

Ans. : Step 1 : Download puttygen for creating a .ppk file as putty doesn't accept .pem file generated by AWS.

Step 2 : Convert your .pem file to .ppk file using PuttyGen. Load your .pem file generated by AWS. Then save the private key (.ppk) file.

Step 3 : Open Putty. Add you IP. Add user name.Add ppk file. Click Open. Give the ip address or the host name. Then in data section give the User name for the instance for linux its generally "ec2-user".

Step 4 : Click open.

Thus now EC2 instance is connected.

6.3 : AWS Elastic Load Balancer and its Types

Q.9 What is elastic load balancer and explain its working.

[SPPU : June-22, Marks 5]

Ans. :
- Every application needs good efficiency, performance and availability.
- The network traffic and load on a single server decrease the efficiency, performance, and availability of an application.
- Elastic load balancer or ELB is a service provided by AWS to distribute incoming traffic across multiple servers or clusters.
- The ELB increases the availability and fault tolerance of an application.
- AWS load balancer will distribute your workloads across multiple compute resources, such as a Virtual Machine or Virtual Server.
- With AWS management console we can create load balancers.

ELB Work

- ELB accepts all the traffic from the client and then routes this traffic to the target that the user wants.
- If the load balancer finds an unhealthy target, then it will stop redirecting its users there and will move with the other healthy targets until that target is declared healthy.

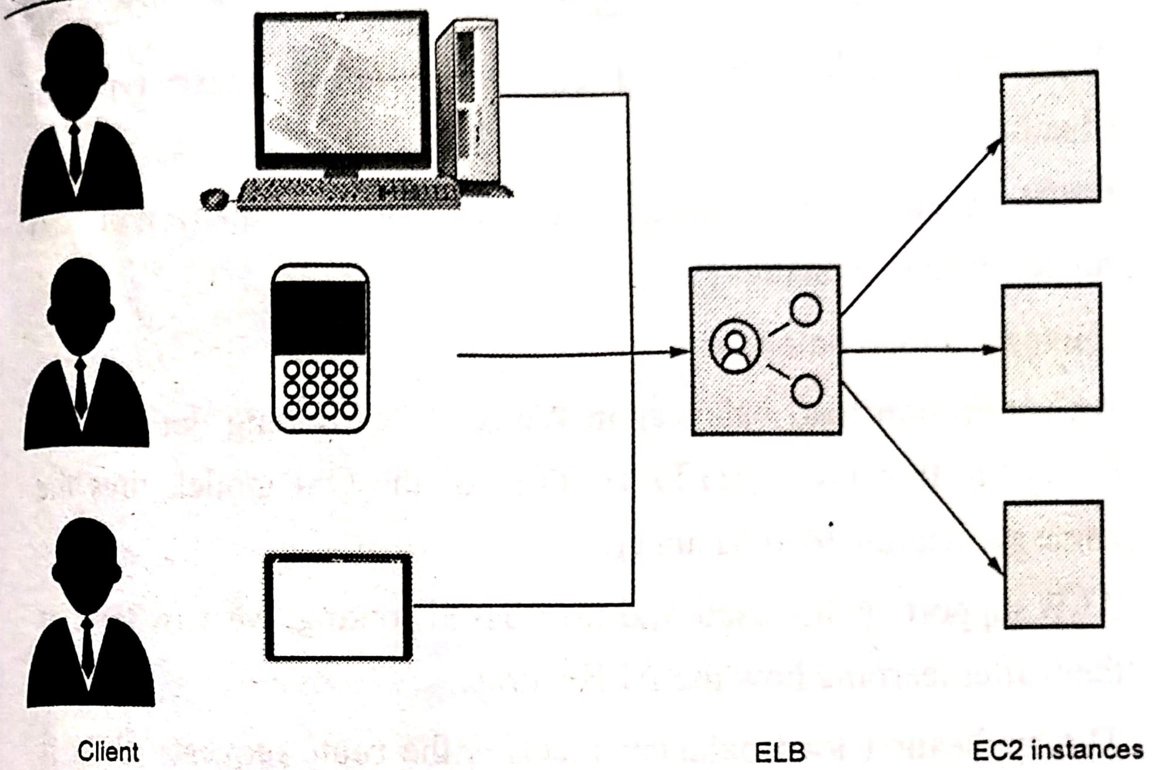

Fig. Q.9.1 ELB

Q.10 Explain in detail types of elastic load balancer.

[SPPU : June-22, Marks 6]

Ans. : 1. Classic load balancer

- It is a traditional load balancer that is used initially.
- The classic Load balancer in AWS is used on EC2-classic instances.
- This is the previous generation's load balancer and also it doesn't allow host-based or path-based routing.
- It ends up reducing efficiency and performance in certain situations.
- It is operated on connection level as well as request level.

2. Network load balancer

- Network load balancer in AWS takes routing decisions in the transport layer (TCP/SSL) of the OSI model.
- It is mainly used for load balancing TCP traffic.
- It can handle millions of requests per second.

- AWS network load balancer can be trusted in these types of situations.
- Widely used to load balancing the TCP traffic and it will also support elastic or static IP.

3. Application load balancer

- An application load balancer in AWS makes routing decisions at the application layer (HTTP/HTTPS) of the OSI model, thus the name application load balancer.
- ALB supports path-based and host-based routing, we will look at them after learning how the ALB works.
- The application load balancer receives the route requests, then it inspects the received packets.
- Then it chooses the best target possible for the type of load and sends it to the target with the highest efficiency.

6.4 : AWS VPC and Component of VPC

Q.11 What is AWS VPC ? **[SPPU : Dec.-22, Marks 3]**

Ans. : • AWS provides security to the cloud and servers with its services.

- VPC or Virtual Private Cloud provides additional security levels on the AWS services that you are using.
- VPS gives you full control over routing traffic to and from your instances.
- AWS VPC is a private subsection of AWS in which you can place AWS resources such as EC2 instances and databases.
- It gives all the benefits of the traditional network that you have for your own data center.
- Resources and applications are accessed through IPv4 or IPv6 in your AWS VPC.

- There are two types of VPC :

  1. Default VPC 2. User defined VPC

Q.12 What are the advantages of VPC ?

Ans. : • EC2 Instance security group membership can be changed while it is running.

- Static IPv4 assigned to instances that persist across the start and stop.
- Access Control List (ACL) is an additional security layer to protect instances.
- Multiple IPv4 can be assigned to your instances.
- Control both inbound and outbound traffic of instances.

Q.13 What are the different components of VPC ?

☞ [SPPU : June-22, Marks 5, Dec.-22, Marks 3]

Ans. : • **Subnet :** A segment of a VPC's where you can place groups to isolated resources.

- **Internet gateway :** VPC side of a connection to utilize public Internet.
- **NAT gateway :** A highly available, managed Network Address Translation (NAT) service for your resources in a private subnet to access the Internet.
- **Virtual private gateway :** The Amazon VPC side of a VPN connection for secure transactions.
- **Peering connection :** To route traffic via private IP addresses between two peered VPCs.
- **VPC endpoints :** Enables private connectivity for your service in AWS without using an Internet Gateway, VPN, Network Address Translation (NAT) devices, or firewall proxies.
- **Egress-only internet gateway :** A stateful gateway that provides egress-only access for IPv6 traffic from the VPC to the Internet.

6.5 : AWS Storage

Q.14 Explain the concept of AWS storage

Ans. : • Cloud storage is a cloud computing model that stores data on the Internet through a cloud computing provider that manages and operates data storage as a service.

- It is delivered on-demand and manages your own data storage infrastructure.
- It is cost-effective for storage buying.
- AWS storage services have different provisions for highly confidential data, frequently accessed data and not so frequently accessed data.

Q.15 Explain any three AWS storage services.

[SPPU : June-22, Dec.-22, Marks 6]

Ans. : 1. Simple Storage Service (S3)

- It is an object storage type and stores any type or size data.
- It is used for web applications, mobile applications, analytics, and backup services.
- It provides user management control for any specific requirement in the application to store the data.
- With S3 we can create, rename, delete folders with the help of a web-based file explorer.
- AWS provides 99.99999999% durability to deliver data to end-users.
- It provides 3 forms of encryption, including server-side encryption and client-side encryption.

2. Amazon S3 Glacier

- It is an object storage type.
- It is used for long-term data storage and especially for archival data.

- It also provides encryption on data for security.
- This service is used for a low retrieval rate of data in any application.
- It allows running queries and analytics on it directly.
- It provides 99 % durability.

3. Elastic Block Storage (EBS)

- It is a block storage type.
- It is like hard drive storage.
- This storage is attached to the EC2 instance and used as block storage, where we can install any operating system.
- It is available in SSD or HDD format.
- EBS volumes are network file systems.
- These volumes get automatically replicated within availability zones for better availability and durability.
- We can dynamically increase or decrease the capacity of storage.

4. Amazon EC2 Instance Storage

- It is a block storage type.
- It is used as temporary storage for EC2 instances.
- The temporary storage of data that changes frequently like buffers, queue cache, etc.
- It uses SSD for high I/O performance.
- Durability provides with replication of storage.

Q.16 What is S3 bucket ?

[SPPU : Dec.-22, Marks 6]

Ans. : • S3 has two primary entities called buckets and objects.

- Objects are stored in buckets.
- Within this bucket, you can create multiple folders to differentiate the files.

- If you want to upload images you may create a folder named images and store it in the logical address of the file with the prefix 'images'
- You can create a maximum of 100 buckets per account.
- The bucket names should be unique.

6.6 : Deploy Website or Web Application on AWS

Q.17 List steps to deploy website on EC2.

Ans. :

Step 1 : Connect server with PuTTY

Step 2 : Update Ubuntu instance

Step 3 : Install Apache 2

Step 4 : Go back to your instances page and click on 'launch-wizard-1' under security groups

Step 5 : Go to Inbound and click on Edit inbound rules

Step 6 : Then, add an HTTP rule with Source as 'Anywhere-IPv4' and save the rule.

Step 7 : On your browser, use your public IP address, put it in the URL box and run

Step 8 : You can create your HTML page or upload your code with GitHub in the/var/www/html folder.

- With these steps, you can deploy the website on the EC2 instance.

Q.18 How to update the Ubuntu server ?

Ans. : To update the Ubuntu server following command is used

sudo apt-get update

Q.19 What is the default port no for HTTP protocol ?

Ans. : The default HTTP and HTTPS ports for the Web server are port 80 and 443, respectively.

6.7 : Launch an Application with AWS Elastic Beanstalk

Q.20 What is elastic beanstalk and enlist the advantages of using it ? [SPPU : June-22, Marks 2]

Ans. : • Elastic Beanstalk is a compute service for web applications.

- It is a pre-configured EC2 server where environment configuration is already done.
- EC2 is infrastructure-as-a-service whereas Elastic Beanstalk is platform-as-a-service.
- It makes it easier for developers to deploy and manage applications on AWS.
- In EC2 we have to create a deployment environment for application deployment like for PHP, Python, Node.JS, etc. but in Elastic Beanstalk, it provides configured server.
- It cost pay-per-use of resources so there is no additional cost for you.

AWS Elastic Beanstalk Advantages

- Developer productivity
- Cost-effective
- Easy to start
- Customization
- Management and updates

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